Yousef Kamel

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Education

- **University of Waterloo** Waterloo, Ontario
  Bachelor of Applied Science in Electrical Engineering; GPA: 4.00 *Expected 2030*

Experience

- **Software & Automation Developer Co-op** North York, Ontario
  Renellence Inc *August 2026 – December 2026*
  - Automated the order-to-cash pipeline end to end in n8n, covering order intake, human-in-the-loop approvals, e-signature gating, QuickBooks Online invoicing and 3PL shipment tracking synced back to HubSpot
  - Built and shipped **Renellence Edge**, a sales operations platform used daily by a five-rep national sales team, with pipeline forecasting, KPI scorecards, a postal-code territory map and event ROI across 20+ screens
  - Architected Edge's serverless Node.js API on Vercel (12 routes, 15K+ lines) with role-based access control and HMAC-signed sessions, replacing N+1 CRM lookups with batched HubSpot v3/v4 calls and a Redis-cached order book
  - Integrated nine external APIs (HubSpot, Stripe, Shopify, QuickBooks, Meta Marketing, SignatureAPI, Google Sheets, VoIP, Gemini) through signed webhooks and token auth, including two-way contact and call sync
  - Designed for failure with paginated batch reads, exponential backoff on rate limits, deduplication and two-phase syncs that never delete before a copy is confirmed, so outages never blank the UI or lose records
  - Wrote a dependency-free test harness (1,600+ assertions, 46 suites) and a mocked local environment simulating every upstream API; released through a scripted dry run, backup and verification against live executions
- **Software Developer Intern** Waterloo, Ontario
  Normative *January 2026 – April 2026*
  - Built **NIMBUS**, a full-stack profitability modeling tool (React, TypeScript, Node.js, PostgreSQL) with a reactive calculation engine for live COGS, margin and profit tracking without unintended re-renders
  - Automated seniority-to-cost mapping from backend rate card data, eliminating manual rate-mapping errors and keeping cost and billing data consistent
  - Built 15+ reusable React, TypeScript and MUI components, eliminating recurring UI-state bugs and manual state-sync workarounds
  - Integrated type-safe tRPC APIs across 10+ endpoints, driving runtime type errors on new API calls to zero
  - Designed PostgreSQL schemas and optimized queries through Prisma ORM, reducing redundant data by 25% and improving query response times by 30%

Projects

- **Patient Motion Detection System** | *C++, ESP32, Python, Bluetooth, Signal Processing* Sep 2025 – Jan 2026
  - Designed an ESP32-based embedded system that classifies patient movement in real time from IMU accelerometer and gyroscope data to flag fall-risk scenarios
  - Developed signal processing and motion classification algorithms in C++ to detect abnormal movement patterns associated with delirium

Technical Skills

- **Languages**: Python, TypeScript, JavaScript, C++, Java, Kotlin, SQL, Verilog, HTML/CSS
- **Frameworks & Libraries**: React, Next.js, Node.js, tRPC, Prisma, Material UI, Tailwind CSS, LangChain, LangGraph
- **Infrastructure & Tools**: PostgreSQL, Redis, Vercel, n8n, REST, GraphQL, Webhooks, Git, GitHub Actions, Docker